

Ejercicio 1

Tabla de Subneteo

Red	Host	Masc	Subnet	Gateway	Alt. IP	Broadcast
admin	45	126	10.24.0.0	10.24.0.1	..2.62	..1.63
ventas	30	127	10.24.0.64	10.24.0.65	..94	..95
soporte	14	128	10.24.0.96	10.24.0.97	..110	..111
defensa	10	128	10.24.0.112	10.24.0.113	..126	..127
R1-R2	2	130	10.24.0.128	10.24.0.129	..130	..131
R2-R3	2	130	10.24.0.132	10.24.0.133	..134	..135
Pc 1			IP	Gateway		
Pc 2			0.2	0.1		
Pc 3			0.64	0.65		
Pc 4			0.98	0.97		
			0.114	0.113		

red

ip
red+1

011: ip

broad-1

broad

red-1

42

Carnet 000 92424

A 00

B 24

Red asonada 10.24.0.0

admin 45

ventas 30

soporte 19

Gerencia 10

R1-R2 2

R2-R3 2

admin

2⁵-2 7/45

255. 255. 255. 11000000

2⁶-2 7/62

192

32-6=26

256-192=64

ventas

2⁵-2 7/30

255. 255. 255. 11100000

224

32-5=27

32

Soporte y Gerencia

2⁴-2 7/19

255. 255. 255. 11110000

240

32-4=28

16

~~Gerencia~~ enlaces r-r2²-2

255. 255. 255. 11111100

255

32-2=30

4



Configuración IP de cada PC.

```
PC1> ip 10.24.0.2 /26 10.24.0.1
Checking for duplicate address...
savePC1 : 10.24.0.2 255.255.255.192 gateway 10.24.0.1

PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> ip 10.24.0.66 /27 10.24.0.65
Checking for duplicate address...
PC1 : 10.24.0.66 255.255.255.224 gateway 10.24.0.65

PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC3> ip 10.24.0.98 /28 10.24.0.97
Checking for duplicate address...
PC1 : 10.24.0.98 255.255.255.240 gateway 10.24.0.97

PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> ip 10.24.0.114 /28 10.24.0.113
Checking for duplicate address...
PC1 : 10.24.0.114 255.255.255.240 gateway 10.24.0.113

PC4> save
Saving startup configuration to startup.vpc
. done
```

Configuración de interfaces de los routers.

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.24.0.1	YES	manual	up	up
FastEthernet0/1	10.24.0.65	YES	manual	up	up
Serial1/0	10.24.0.129	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    10.24.0.97      YES manual up              up
FastEthernet0/1    unassigned      YES unset  administratively down down
Serial1/0          10.24.0.130     YES manual up              up
Serial1/1          10.24.0.133     YES manual up              up
Serial1/2          unassigned      YES unset  administratively down down
Serial1/3          unassigned      YES unset  administratively down down

R3#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    10.24.0.113     YES manual up              up
FastEthernet0/1    unassigned      YES unset  administratively down down
Serial1/0          unassigned      YES unset  administratively down down
Serial1/1          10.24.0.134     YES manual up              up
Serial1/2          unassigned      YES unset  administratively down down
Serial1/3          unassigned      YES unset  administratively down down
R3#
```

Tabla de rutas estáticas configuradas.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C       10.24.0.0/26 is directly connected, FastEthernet0/0
C       10.24.0.64/27 is directly connected, FastEthernet0/1
S       10.24.0.112/28 [1/0] via 10.24.0.130
S       10.24.0.96/28 [1/0] via 10.24.0.130
C       10.24.0.128/30 is directly connected, Serial1/0
S       10.24.0.132/30 [1/0] via 10.24.0.130
```

```
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

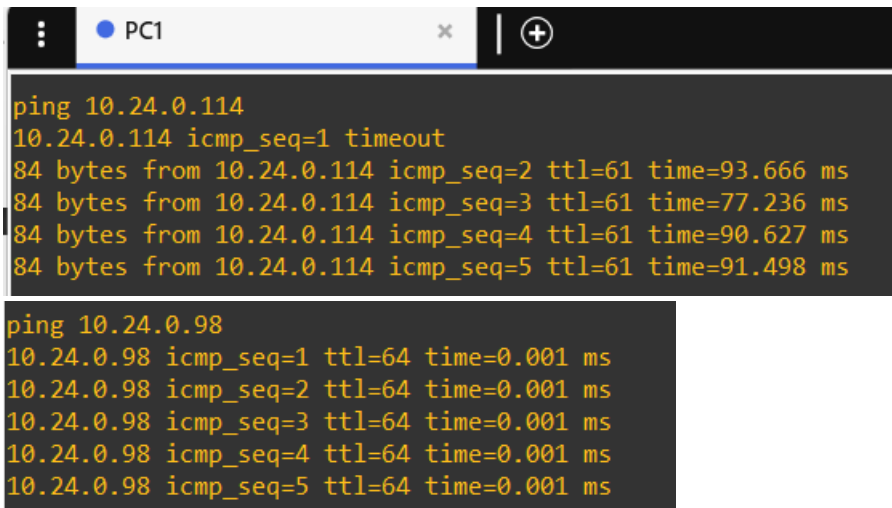
```
    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.129
S       10.24.0.64/27 [1/0] via 10.24.0.129
S       10.24.0.112/28 [1/0] via 10.24.0.134
C       10.24.0.96/28 is directly connected, FastEthernet0/0
C       10.24.0.128/30 is directly connected, Serial1/0
C       10.24.0.132/30 is directly connected, Serial1/1
```

```
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.133
S       10.24.0.64/27 [1/0] via 10.24.0.133
C       10.24.0.112/28 is directly connected, FastEthernet0/0
S       10.24.0.96/28 [1/0] via 10.24.0.133
S       10.24.0.128/30 [1/0] via 10.24.0.133
C       10.24.0.132/30 is directly connected, Serial1/1
```

- Pruebas de ping exitosas entre PC1, PC2, PC3 y PC4



```
ping 10.24.0.114
10.24.0.114 icmp_seq=1 timeout
84 bytes from 10.24.0.114 icmp_seq=2 ttl=61 time=93.666 ms
84 bytes from 10.24.0.114 icmp_seq=3 ttl=61 time=77.236 ms
84 bytes from 10.24.0.114 icmp_seq=4 ttl=61 time=90.627 ms
84 bytes from 10.24.0.114 icmp_seq=5 ttl=61 time=91.498 ms

ping 10.24.0.98
10.24.0.98 icmp_seq=1 ttl=64 time=0.001 ms
10.24.0.98 icmp_seq=2 ttl=64 time=0.001 ms
10.24.0.98 icmp_seq=3 ttl=64 time=0.001 ms
10.24.0.98 icmp_seq=4 ttl=64 time=0.001 ms
10.24.0.98 icmp_seq=5 ttl=64 time=0.001 ms
```

Ejercicio 2

carpet 00042424

A 00

B 24

Red base asignada 192.24.0.0

Pc1 40

Pc2 28

Pc3 18

Pc4 9

Pc5 5

R1-R2

R1-R3

R2-R5

R3-R4

R4-R5

2

2

2

2

2

1)

$2^6 - 27/40$

255.255.255.

11000000

192

$32 - 6 = 26$

64

$w = .63$

2) y 3)

$2^5 - 27/28$

255.255.255.

11100000

$2^5 - 27/18$

224

$32 - 5 = 27$

32

$32 - 5 = 27$

$w = .31$

4)

$2^4 - 27/9$

255.255.255

11110000

240

$32 - 4 = 28$

16

$w = .15$

5)

$2^3 - 27/5$

255.255.255.

11111000

248

$32 - 3 = 29$

8

$w = .7$

replaces

$2^2 - 27/2$

255.255.255.

11111100

252

$32 - 2 = 30$

4

$w = .3$

VLSM	Host	Masc	OSPF	IP	Gate	OL, 30	brodd
Rd	40	126	0.0.0.63	192.24.0.0	192.24.0.1	.62	.63
Rc1	28	127	0.0.0.31	192.24.0.64	19.24.0.65	.94	.95
Rc2	18	127	0.0.0.31	..96	..97	.126	.127
Rc3	9	128	0.0.0.15	..128	..129	.142	.143
Rc4	5	129	0.0.0.7	..144	..145	.150	.151
R1-R2	2	130	0.0.0.3	..152	..153	.154	.155
R4-R3	2	130	0.0.0.3	..156	..157	.158	.159
R2-R5	2	130	..3	..160	..161	.162	.163
R3-R4	2	130	..3	..164	..165	.166	.167
R4-R5	2	130	..3	..168	..169	.170	.171
				..172			

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.4.0.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial1/0	192.4.0.153	YES	manual	up	down
Serial1/1	192.24.0.157	YES	manual	up	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down

```
R1#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
        E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
```

```
        ia - IS-IS inter area, * - candidate default, U - per-user static route
```

```
        o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    192.4.0.0/26 is subnetted, 1 subnets
```

```
C       192.4.0.0 is directly connected, FastEthernet0/0
```

```

R2#show ip
*Mar  1 00:02:14.659: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to down
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.65     YES manual up          up
FastEthernet0/1  unassigned      YES unset  administratively down down
Serial1/0        192.24.0.154    YES manual up          up
Serial1/1        unassigned      YES unset  administratively down down
Serial1/2        192.24.0.161    YES manual up          down
Serial1/3        unassigned      YES unset  administratively down down
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.24.0.152/30 is directly connected, Serial1/0
C       192.24.0.64/27 is directly connected, FastEthernet0/0

```

```

R3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.97     YES manual up          up
FastEthernet0/1  unassigned      YES unset  administratively down down
Serial1/0        unassigned      YES unset  administratively down down
Serial1/1        192.24.0.158    YES manual up          up
Serial1/2        unassigned      YES unset  administratively down down
Serial1/3        192.24.0.165    YES manual up          up
R3#show ip
*Mar  1 00:02:33.899: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.157 on Serial1/1 from LOADING to FULL, Loading Done
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.24.0.156/30 is directly connected, Serial1/1
C       192.24.0.164/30 is directly connected, Serial1/3
C       192.24.0.96/27 is directly connected, FastEthernet0/0

```

```
R4#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.24.0.145	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial1/0	192.24.0.169	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	192.24.0.166	YES	manual	up	up

```
R4#show ip route
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.0.0/24 is variably subnetted, 3 subnets, 2 masks

C 192.24.0.144/29 is directly connected, FastEthernet0/0

C 192.24.0.168/30 is directly connected, Serial1/0

C 192.24.0.164/30 is directly connected, Serial1/3

```
R4#
```

```
*Mar 1 00:03:03.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
```

```
*Mar 1 00:03:03.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
```

```
R5#s
```

```
*Mar 1 00:03:14.047: %OSPF-5-ADJCHG: Process 1, Nbr 192.24.0.169 on Serial1/0 from LOADING to FULL, Loading Done
```

```
R5#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.24.0.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial1/0	192.24.0.170	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	192.24.0.162	YES	manual	up	up
Serial1/3	unassigned	YES	unset	administratively down	down

```
R5#show ip route
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.24.0.0/24 is variably subnetted, 4 subnets, 3 masks

O 192.24.0.144/29 [110/74] via 192.24.0.169, 00:00:04, Serial1/0

C 192.24.0.168/30 is directly connected, Serial1/0

C 192.24.0.160/30 is directly connected, Serial1/2

C 192.24.0.0/26 is directly connected, FastEthernet0/0